
Algorithm 3 Inverse Kinematics obtained by optimization $\mathbf{u}_a = f^{-1}(\mathbf{y}_e)$

- 1: Initialize $\mathbf{q}^0 = [\mathbf{q}_1^0; \mathbf{q}_2^0; \mathbf{q}_3^0; \mathbf{q}_4^0]$
- 2: **for** $i = 1, \dots, \text{maxiter}$ **do**
- 3: Form the residual: ▷ simulate states and gradients

$$\begin{cases} \mathbf{b}_1 \leftarrow \mathbf{F}_1(\mathbf{q}_1^{i-1}) + \mathbf{M}_1 \mathbf{g} \\ \qquad \qquad \qquad \vdots \\ \mathbf{b}_4 \leftarrow \mathbf{F}_4(\mathbf{q}_4^{i-1}) + \mathbf{M}_4 \mathbf{g} \end{cases}$$

- 4: Compute $\mathbf{A}_j(\mathbf{q}_j^{i-1})$, $\mathbf{H}_j$ ($j \in [1, 4]$), $\mathbf{A}$, and $\mathbf{H}_a, \mathbf{H}_e, \delta_a(\mathbf{q}^{i-1})$
- 5: Find $d\mathbf{q}$: ▷ QP solve

$$\begin{cases} \min_{\lambda_a, d\mathbf{q}} \frac{1}{2} \|\delta_e(\mathbf{q}_{i-1}) + \mathbf{H}_e d\mathbf{q} - \mathbf{y}_e\|^2 \\ \text{s.t. } \mathbf{A} d\mathbf{q} \end{cases} = \mathbf{b} + \mathbf{H}_e^\top \lambda_e + \mathbf{H}_a^\top \lambda_a \quad (20)$$

- 6: $\mathbf{q}^i \leftarrow \mathbf{q}^{i-1} + \alpha d\mathbf{q}$ ▷ take solver step
 - 7: **if** $\|\mathbf{q}^i - \mathbf{q}^{i-1}\| < \epsilon$ **then**
 - 8: **return** : $\mathbf{u}_a = \delta_a(\mathbf{q}^{i-1}) + \mathbf{H}_a d\mathbf{q}$ ▷ solver converged
 - 9: **end if**
 - 10: **end for**
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